

JANHAVI BAWANKULE

Final Year B.Tech | Electronic & Telecommunication Engineering Student | Honours in ETC | Embedded System | IOT

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PROFESSIONAL SUMMARY

Electronic & Telecommunication undergraduate (B.Tech – (B.Tech Honours)), St. Vincent Palloti College of Engineering and Technology, Nagpur) combining hands-on experience in Embedded Systems, IoT, Artificial Intelligence, and Telecommunications through academic projects, industrial internships, and research. Proficient in C, C++, Python, MATLAB, Arduino, HDL and embedded programming, with practical experience in AI-based IoT applications, sensor integration, and telecom technologies. Authored an IEEE-format research paper on AI-powered EV infrastructure integrating AI, IoT, GIS, and cloud computing, and completed industrial internships spanning GIS/remote sensing, telecom networking, and manufacturing quality assurance. Passionate about solving real-world engineering challenges through innovation, continuous learning, and collaborative problem-solving, while seeking an entry-level engineering role to create meaningful impact.

KEY HIGHLIGHTS

- Authored an IEEE-format research paper on AI-powered EV Smart Infrastructure, integrating AI, GIS, IoT, and Cloud Computing.
- Completed three industrial internships spanning GIS & Remote Sensing (MRSAC), Telecom Networking (BSNL), and Manufacturing Quality Assurance (Subros Limited).
- Independently designed & develop a café website using Replit, implementing responsive design and interactive front-end features.
- Serving as a Career Development Cell (CDC) Coordinator, coordinating placement activities, student engagement, and career development initiatives.
- Qualified for Round 1 of the Google Student Ambassador and Gemini Student Ambassador Programs (2026).

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, basic of Java
- **Embedded Systems:** Arduino, 8051 Microcontroller, ATmega Microcontroller, 8086 Microprocessor, Embedded C, HDL.
- **Software & Development Tools:** MATLAB, Multisim, Keil µVision, MASM, Tinkercad, Xilinx ISE, Replit.
- **Research & Emerging Technologies:** Artificial Intelligence (AI), Internet of Things (IoT), Geographic Information Systems (GIS).
- **Hardware & Design:** PCB Design, Circuit Design & Simulation .
- **Core Skills:** Problem-Solving, Debugging, Hardware–Software Integration, Analytical Thinking, Team Collaboration.

PROJECTS

6 – Wheel Rover with Robotic Arm & Camera | Team Project

- Designed and developed a 6-wheel robotic rover integrated with a robotic arm and camera for remote navigation and object handling.
- Implemented real-time monitoring and obstacle detection using embedded systems and sensor integration for surveillance and automation applications.
- Enhanced mobility across diverse terrains, enabling use cases in search & rescue, agriculture, and industrial inspection.

AI-Driven Fish Farm Monitoring System | Team Project

- Engineered an AI-powered IoT monitoring system using embedded systems and sensors to continuously track water quality parameters, including temperature, TDS, and light intensity.
- Implemented AI-driven analysis and intelligent alert notifications for abnormal water conditions, supporting predictive monitoring and efficient fish farm management.

Smart Hay Cutter | Independent Project

- Engineered an embedded-C agricultural automation vehicle on Arduino Uno to reduce manual farm labor, adding an automated seed-sowing mechanism and a fire-extinguishing function.
- Collaborated with a project team to design, build, and test the full hardware-software integration.

Intelligent Digital Queue & Smart calling management System | Team Project

- Designed and developed an embedded solution for automated queue management and real-time digital call display, enhancing operational efficiency and minimizing manual intervention.
- Implemented intelligent notification, display interfacing, and microcontroller-based control to provide a scalable and reliable communication system for institutional and healthcare applications.

My Café – Responsive Frontend Web Application | Independent Project

- Developed a responsive café website using with an intuitive and mobile-friendly user interface.
- Leveraged Replit AI-assisted development environment to accelerate development, testing, and deployment while following modern frontend design practices.

RESEARCH

EV Smart Platform: Intelligent EV Infrastructure — Alerts, Mapping & Services | IEEE-Format Research Project

- Researched an AI-powered electric vehicle ecosystem integrating GIS-based route optimization, predictive battery health monitoring, and intelligent charging station discovery.
- Explored the integration of AI, IoT, GIS, Cloud Computing, and Intelligent Transportation Systems (ITS) to improve EV infrastructure efficiency and user experience.
- Authored the research paper in IEEE format, covering literature review, system design, technical analysis, and evaluation of intelligent transportation solutions.

INDUSTRIAL EXPERIENCE

Industrial Intern — Maharashtra Remote Sensing Applications Centre (MRSAC), Nagpur | [June 2026 – July 2026] · 1 month

- Completed a one-month internship in GIS, Remote Sensing, and Geospatial Technologies, gaining hands-on exposure to spatial data analysis and digital mapping.
- Worked with satellite imagery and geospatial datasets to support infrastructure planning, environmental monitoring, and smart city use cases.

Industrial Intern — Bharat Sanchar Nigam Limited (BSNL) | [May 2025 – June 2025]

- Completed industrial training across National Internet Backbone (NIB), Fiber-to-the-Home (FTTH), Routing & Networking, and Sales & Marketing departments.
- Gained practical exposure to optical fiber communication, broadband technologies, router configuration, and network architecture.

Diploma Engineer Trainee (DET) — Subros Limited, Pune | [Jan 2024 – May 2024] · 6 months

- Completed a six-month industrial training program in automotive air-conditioning manufacturing, working across Quality Control, HVAC Assembly, Condenser and Production Operations.
- Performed quality inspections and component validation, ensuring compliance with manufacturing quality standards and safety procedures.
- Gained hands-on experience in quality assurance, process optimization, and cross-functional team collaboration within a high-volume manufacturing environment.

LEADERSHIP & EXPERIENCE

- **CDC Coordinator** – Coordinated placement-related activities and organized multiple technical and professional events, including debates and student competitions.
- **Core Organizer, Festronix** – Planned and managed technical events, including LogicQuest, collaborating with teams to ensure successful event execution and participant engagement

EDUCATION

- **B.Tech, Electronic & Telecommunication Engineering** — St. Vincent Pallotti College of Engineering and Technology, Nagpur (In progress; 7 SGPA through 6th semester)
- **B.Tech (Honours) in Electronic & Telecommunication.**
- **Diploma, Electronics & Telecommunication Engineering** — Government Polytechnic, Nagpur (2024; 79.15 %)
- **Xth** — New English High School, Congress nagar Nagpur (2021; 86.20 %)

CERTIFICATIONS

- NPTEL (5 Courses)
- Crest CTF 2026
- MATLAB Onramp
- Microchip EduSkills – VLSI
- Infosys Springboard certification
- MATLAB Onramp — Self-paced training course
- Inter College Hackathon Certification
- TCS-iON Young Professional Course